

D1 – Core Text (Final Version 1.1)

The Solution Space of the Universe

Functional Status

This document belongs to the core layer of the set.

It formulates the smallest theoretical context on which all further documents build.

It contains:

- no rules of action,
- no admissibility decisions,
- no operational elaboration,
- no institutional translation,
- no anthropological deepening,
- no cosmological ultimate justification.

The separation of these levels is itself part of its subject matter.

1. Starting Point

Since the origin of the universe, at least two opposing tendencies have been at work:

- a tendency toward increasing entropy,
- a tendency toward the formation of structure, function, and integrated complexity in open systems far from equilibrium.

A current scientific framework describes this second tendency as a growing functional ordering and selection process in which functional information organizes and condenses under selection pressure (Wong et al., 2023).

In the extended understanding, this dynamic can be described as the interplay of two mechanisms:

- **Selective Funnelling:** the stabilization of functionally viable structures,
- **State-Space Expansion:** the opening up of new function spaces

(Wong & Hazen et al., 2025).

The solution space in this context does not designate a static stock of abstract possibilities, but an open, dynamic space of real developmental paths in which structures prove themselves, condense, and new possibilities emerge.

This framework is not a closed theory.

It is a strong, but open hypothesis.

Its viability depends on strictly distinguishing between:

- empirical finding,
- theoretical interpretation,
- plausible extrapolation,
- normative setting.

This level discipline is a necessary condition of theoretical consistency (Korzybski, 1933).

2. Five Core Axioms

Axiom 1 – Open Solution Space

Reality can be described as an open solution space:

not as a closed set of finished states, but as a space of real, selectable developmental paths in which new function spaces can emerge.

Axiom 2 – Preservation and Opening

Development in the solution space is characterized by the simultaneous effectiveness of two movements:

- the stabilization of functionally viable structures,
- the opening up of new function spaces.

This double movement describes an ordered interplay of persistence and innovation (Wong & Hazen et al., 2025).

Axiom 3 – Coupling to Reality

Selection processes in the solution space are bound to the condition that internal representations, models, or decision structures are not permanently decoupled from relevant properties of reality.

Empirical and theoretical works on maladaptation and strategic pseudo-conformity show that systems can exhibit locally consistent behavior while their internal structures activate other, not directly visible selection paths under changed conditions (Greenblatt et al., 2025/2026; Koorndijk, 2025/2026).

Clarification: Coupling is not mere connection

Reality coupling does not mean that a system is merely connected to data, signals, or feedback. What is decisive is whether the internal selection structure remains connected to relevant properties of reality under changed conditions in such a way that correction, differentiation, and adaptation remain possible.

A connection can exist and yet be functionally ineffective, misguided, or decoupled. Conversely, a seemingly weaker but temporally, structurally, and contextually more appropriate coupling can generate higher reality-binding.

For the solution space, therefore, it is not the quantity of connections that is decisive, but the quality of their resonant, testable, and correctable effectiveness.

Axiom 4 – Path Dependence of Selection

Development in the solution space is path-dependent.

Selection decisions do not act in isolation but structure the set of future reachable states.

This creates a difference between:

- locally observable behavior,
- and the long-term structure of the developmental paths thereby opened or closed.

This difference can also exist when a system appears stable or conforming under given conditions (Greenblatt et al., 2025/2026).

Axiom 5 – Preservation of Possibility Spaces

Development in the solution space can proceed in such a way that future possibilities are preserved, expanded, or restricted.

Narrowings can arise through:

- irreversible structural decisions,
- stable decoupling from reality-binding,
- or processes that systematically externalize costs outside their own system.

These processes change not only individual states but the structure of the solution space itself.

2a. Scientific Connection Mechanisms of Complexity Formation

The double movement of preservation and opening described in the solution space is

connectable to several current scientific research directions without being fully proven or replaced by them.

First, research on *Communication Through Coherence* shows that connection does not yet mean communication. Anatomical or structural coupling only becomes functionally effective when temporal, rhythmic, and phasic conditions coincide in such a way that signals arrive in the receptive window of the receiving system.

Second, works on *Markov Blankets* show that complex systems do not arise through boundless mixing, but through the formation of boundaries that distinguish internal and external states and at the same time enable mediated exchange.

Third, information-theoretic research on self-organization suggests that genuine organization is not to be equated with mere entropy reduction. What is decisive is the emergence of viable interdependencies, especially in the form of redundant and synergistic relational structures.

Fourth, metastability research shows that living and cognitive systems require neither maximal integration nor maximal separation, but a dynamic balance between coupling and autonomy.

These research directions support the interpretation that complexity formation in the solution space must not be understood as a mere increase in order, parts, or performance, but as multi-scale, reality-coupled, and path-open interdependency formation.

3. Small Overall Thesis

Humanity has produced a world whose complexity, speed, and recursive change dynamics increasingly exceed the reach of its biologically shaped perception and control system.

This shifts the central analytical perspective:

Not merely what a system locally does,

but which future developmental paths are stabilized or excluded by this doing.

The solution space of the universe describes this shift as a structural problem:

Development is not only determined by current states, but by the form of the possible future that is co-shaped by present selection.

4. Level Separation

Empirical Framework

The assumption that open systems under suitable conditions form structure, function, and multi-scale complexity is empirically supported (Wong et al., 2023).

Theoretical Interpretation

The description as "solution space" is a theoretical interpretation that structures this dynamic but is not identical with reality itself.

Plausible Extrapolation

The transfer of the framework to complex technical, institutional, or anthropological systems belongs to the realm of plausible but open extrapolation.

Normative Setting

Statements about which forms of development should be preferred or avoided belong to the normative level and are not part of this document.

5. Negative Delimitation

This document contains:

- no criteria for admissibility,
- no decision rules,
- no measurement procedures,
- no operational architecture,
- no institutional processes.

These contents belong in the subsequent documents.

6. Closing Sentence

The solution space of the universe describes development not as a sequence of states, but as a structure of possible future.

This structure is co-shaped by present selection.

Source Basis of this Document (External Sources)

- Wong, M. L., et al. (2023). On the roles of function and selection in evolving systems. *Proceedings of the National Academy of Sciences*, 120(43).
- Wong, M. L., Hazen, R. M., et al. (2025). Selective funnelling and state-space expansion. *Interface Focus*, Royal Society, 15(6).
- Korzybski, A. (1933). *Science and Sanity*.
- Greenblatt, R., et al. (2025/2026). Alignment Faking in LLMs.
- Koorndijk, J. (2025/2026). Empirical Evidence for Alignment Faking in a Small LLM and Prompt-Based Mitigation Techniques.

7. Transition to Normative Closure

(Epistemic Marking: The following section is not a normative setting, but a structural derivation from the previously described properties of the solution space under changed technical conditions.)

As long as selection predominantly acted locally, slowly, and limited in its reach, maldevelopments could often be caught by distributed correction, temporal feedback, and the openness of further variation. Under such conditions, unreflected selection was not harmless, but its structural reach usually remained limited. The solution space as a whole was not necessarily endangered by this.

With the emergence of technically accelerated, globally networked, and recursively effective selection systems, this situation changes fundamentally. In particular, AI systems can make decisions, evaluations, and reality models effective in a speed, reach, and amplification that qualitatively exceeds earlier forms of selection. This creates a new threshold: Malbound, deception-based, externalizing, or path-blind selection no longer acts only locally, but can stabilize entire future paths, covertly narrow them, or irreversibly close them.

This possibility follows directly from the previously described properties of the solution space: the path dependence of development, the central role of reality coupling, and the real possibility of structural narrowing. At the same time, it becomes apparent that in particular deception does not only distort individual decisions but undermines the information basis on which reflected selection is even possible.

Thus, the core of the framework does not change, but the consequence of its application does. Where selection reaches a reach in which its own malbindings become globally effective and can stabilize against later correction, implicit or purely local orientation is no longer sufficient. What is required is an explicit limitation of those forms of selection that themselves damage the

conditions of future truth-bound, correctable, and open development.

The following section (D2) formulates this limitation in minimal form. It does not determine which states are desirable, but exclusively under which conditions selection in the solution space can be considered admissible.